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Deep Learning Approaches for Aviation Weather Forecasting

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Abstract

Weather forecasting is critical for aviation safety, with weather-related incidents accounting for 10% of aviation accidents. This research evaluates deep learning approaches for predicting meteorological conditions at Wellington International Airport using historical METAR data from 2020-2025. We compare Artificial Neural Networks (ANN) with backpropagation and Long Short-Term Memory (LSTM) networks for forecasting temperature, wind speed, visibility, and cloud cover at a 1-hour interval. Results demonstrate that LSTM models consistently outperform ANN architectures across regression tasks, achieving MAE improvements of 3.56 knots for wind speed, 1.16 SM for visibility, and 4.71°F for temperature. For cloud cover classification, both models achieved approximately 54% accuracy across eight categories. These findings suggest that deep learning models, particularly LSTMs, can effectively leverage temporal patterns in meteorological data to provide accurate short-term forecasts while requiring significantly less computational resources than traditional Numerical Weather Prediction methods.

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Chapter 1

Introduction

Weather forecasting is one of the most important factors in aviation travel. Weather-related incidents in air travel account for 10% of cases, being in the top three behind pilot error and mechanical failure [1]. Weather reports are used to help aid pilots with decisions around flight planning like METARs and TAFs. METARs or Meteorological Aerodrome Reports are recorded weather observations made every 30 minutes to assist pilots with current meteorological conditions. TAFs or Terminal Area Forecasts are the standard forecasting reports that pilots use. TAFs are made valid for a 24-hour period and issued four times daily at 12am, 6am, 12pm and 6pm.

1.1 Motivation

The problem in this paper refers to the need to have forecasting systems that reduce the error from predictions of meteorological features like temperature, wind, and visibility.

Current NWP (Numerical Weather Prediction) systems are used to help forecast weather. NWP utilises complex mathematical modelling of current state to predict the future state of the atmosphere using data from observations of its current state [2]. This is the basis of TAFs used at aerodromes to forecast weather. With the advent of deep learning models like the Neural Network and transformer presents a unique opportunity to compare discriminative ability when forecasting meteorological conditions at specific time intervals against current NWP methods. Utilisation of deep learning models also hold unique value in the enablement of combining them as ensembles at low computational cost and challenging the inherent model biases that are present in NWP models [3].

These traditional NWP methods also run on HPC (High Performance Computing) systems which have massive environmental impacts. Its rapid advancement presents real environmental challenges [4].

We want to leverage deep learning methods in machine learning to see how the discriminative ability of models like an LSTM and Artificial Neural Network with backpropagation perform on predicting features relating to weather such as temperature, wind speed and cloud cover. These predictions pertaining to weather conditions at aerodromes can be compared to traditional forecasting methods like TAFs (Terminal Aerodrome Forecast).

1.2 Background

Pilots before flight carry out important inspections, mandatory checks and analysis of weather for departure procedures and onset of flight. Pilots are provided with METAR reports that give present time information of weather. Features such as wind speed/wind direction, visibility, cloud coverage, temperature/dew point and altimeter setting. Certain variables in these reports dictates the flight plan and procedure to ensure flight safety.

This presents the unique opportunity to allow machine learning models to leverage previous METAR reports to forecast meteorological features 1 hour after current weather observations. This could greatly reduce the cost of producing TAFs as local aerodromes have a high upfront cost of installing relevant infrastructure such as AWS (Automated Weather Stations). A report conducted by the Bureau of Meteorology of Australia outlined that an aerodrome could expect a cost of $100,000 and $150,000 to install an AWS [5].

The airport chosen to conduct this research was Wellington International Airport (NZWN). Wellington is one of the few cities in the world that is most notorious for its severe wind and weather events. This has led to a genuine fear of flying being unravelled among passengers travelling to the capital, with scientists from the National Institute of Water and Atmospheric Research (NIWA) outlining that the worst time to fly into Wellington is in the springtime [6]. Low visibility events such as fog is also a common occurrence which makes Wellington International Airport an interesting airport to research.

Chapter 2

Literature Review

2.1 History of Weather Prediction

NWP remains the traditional approach to weather forecasting for aviation procedures and operations. TAFs (Terminal Area Forecast) offer detailed forecasts for the next 24 to 30 hours. TAFs are produced by trained meteorologists using numerical weather prediction (NWP) models, which work by solving complex differential equations representing atmospheric physics [2].

NWP models are physically grounded but still present numerous limitations such as model bias [3] and use of intense computational resources [4]. NWP models often struggle with local phenomena and require extensive infrastructure in the form of Automated Weather Stations (AWS) for initialisation and validation. The high upfront cost of installing and maintaining AWS infrastructure at smaller aerodromes can be prohibitive, leaving many regional airports without dedicated forecasting capabilities [5].

A simpler baseline approach is the persistence model which assumes that weather conditions now will most likely be the same in the future at small intervals. This is computationally trivial, but persistence models often struggle after the fact with limited accuracy for short time horizons. Perez-Ortiz et al. [7] identified that persistence models for visibility prediction achieved an average mean absolute error (AMAE) of 3.17, significantly worse than more sophisticated approaches that we will explore.

2.2 Evolution into Machine Learning Algorithms

With the emergence of machine learning, particularly with deep learning approaches such as the Artificial Neural Network, has opened new possibilities for weather forecasting. Unlike physics-based NWP systems, data-driven approaches can learn complex patterns directly from historical observations without requiring explicit modelling of atmospheric dynamics.

Machine learning offers several advantages for aviation weather forecasting. They require significantly less computational power than NWP models, making them suitable for deployment at resource-constrained regional airports. Additionally, they can leverage the extensive historical METAR (Meteorological Aerodrome Report) data that airports already collect, eliminating the need for additional specialised infrastructure. Most importantly with neural networks they can capture non-linear relationships and local patterns that NWP tend to struggle with.

2.3 Deep Learning for Weather Prediction

2.3.1 Common Methodologies Across Studies

A review of recent literature reveals several consistent methodological choices across deep learning applications to aviation weather forecasting. The studies [7, 8, 9, 10] all utilise METAR reports as the form of data acquisition, demonstrating the value that these have when training ML models.

Common preprocessing approaches across multiple studies [7, 8, 9, 10] use feature selection as a method to focus on the core meteorological variables that have found to be the strongest predictors across different forecasting tasks. Variables such as temperature, humidity, wind speed, and direction, visibility, and pressure help facilitate stronger overall model performance due to its relevance. Temporal features such as hour of day and day of year were also frequently included to capture diurnal and seasonal patterns.

The Adam Optimiser emerged as the standard choice for training deep learning models across studies, with researchers noting its superior performance over traditional Stochastic Gradient Descent [8, 9, 10]. Similarly, Mean Squared Error (MSE) was the dominant loss function for regression tasks, while cross-entropy was used for classification problems.

2.3.2 Model Architecture Comparisons

Recurrent Networks: LSTMs and RNNs

Long Short-Term Memory (LSTM) networks have emerged as a particularly promising architecture for weather forecasting due to their ability to capture temporal dependencies in sequential data. Alves et al. [8] conducted a comprehensive comparison of five deep learning architectures (LSTM, vanilla RNN, 1D-CNN, CNN-LSTM, and GRU) for wind prediction, finding that LSTM achieved the best overall performance with an MAE of 1.23 m/s for wind speed and a circular MAE of 15.80 degrees for wind direction. These results align itself with the acceptable threshold for operational TAF predictions, suggesting LSTMs could meet operational standards.

Similarly, Choi and Kim [10] found that LSTM models outperformed both traditional Multi-Layer Perceptron's (MLPs) and vanilla RNNs for predicting airport arrival and departure capacity based on weather conditions. Their LSTM achieved the lowest test RMSEs across both prediction tasks, ranging from 1.7-3 for departures and 1.84-4.27 for arrivals. The consistency of LSTM superiority across different prediction tasks i.e wind, capacity and different research groups suggests this architecture is particularly well-suited for aviation weather applications.

However, LSTMs are not universally superior. Schultz et al. [9] noted that their LSTM struggled with certain classification tasks, achieving only 48.9% accuracy for predicting airport arrival efficiency categories. This suggests that while LSTMs excel at continuous variable prediction, their advantage may be less pronounced for classification tasks. Additionally, Alves et al. [8] found that 1D-CNNs outperformed LSTMs for short-interval wind direction forecasting, indicating that task-specific architecture selection remains important.

Feed Forward Networks

Despite the advantages of recurrent architectures, simpler feedforward networks have also demonstrated value in certain applications. Araujo and Andrade [11] showed that a simple neural network with three layers and backpropagation could improve the generalisation ability for temperature by 22-28% compared to traditional NWP methods. The RMSE improvement was found to be at 28.75°F which suggests for certain well-behaved variables like temperature, the additional complexity of recurrent architectures may well not be needed.

Perez-Ortiz et al. [7] took a different approach, using an autoregressive neural network within a Mixture of Experts framework (MoE) for visibility prediction. Their STMEIC (Simultaneously Trained Mixture of Experts with Imbalanced Costs) model achieved an AMAE of 1.17 SM for predicting low-visibility events, compared to 3.17 for a persistence model. This shows a 63% improvement, demonstrating that ensemble approaches combining multiple neural networks can be particularly effective for challenging prediction tasks.

Chapter 3

Methodology

3.1 Data Acquisition

The data utilised for the deep learning experiments was the METAR reports made publicly available on the IOWA State MESONET site [14]. Selection of METAR reports were made in a timeframe from 2020 to 2025 at NZWN (Wellington International Airport).

3.2 Data Preprocessing

METAR reports were already converted into a format that was simple to separate in terms of variables with our data frames. The MESONET separates the data into 17 features being:

1. Station Identifier
2. Timestamp
3. Temperature (Fahrenheit)
4. Dew Point
5. Relative Humidity (%)
6. Wind Direction (degrees)
7. Wind Speed (knots)
8. Precipitation
9. Pressure Altimeter
10. Sea Level Pressure
11. Visibility (miles)
12. Wind Gust (knots)
13. Sky Coverage
14. Sky Level
15. Feel (apparent temperature)

16. Ice Accretion
17. Snow Depth

For training our models, four targets were chosen in terms of relevance to previous research carried out in the field [7, 8, 9, 10] and call for future work to explore the temporal patterns. These targets were:

1. Temperature
2. Wind Speed
3. Visibility
4. Cloud Cover

Where the deep learning models predict a continuous value for temperature, wind speed and visibility, whereas cloud cover is split into categories leveraging deep learning models that classify.

3.2.1 Exploratory Data Analysis

The time series dynamics are uncovered during the Exploratory Data Analysis step. Wind speed had the most informative trends with seasonal spikes in the earlier and later parts of the year. Temperature showed a consistent parabolic relationship throughout the years. The earlier parts of the year saw consistent decrease to a stable trough in months 6-8 and then an increasing trend into the later parts of the year.

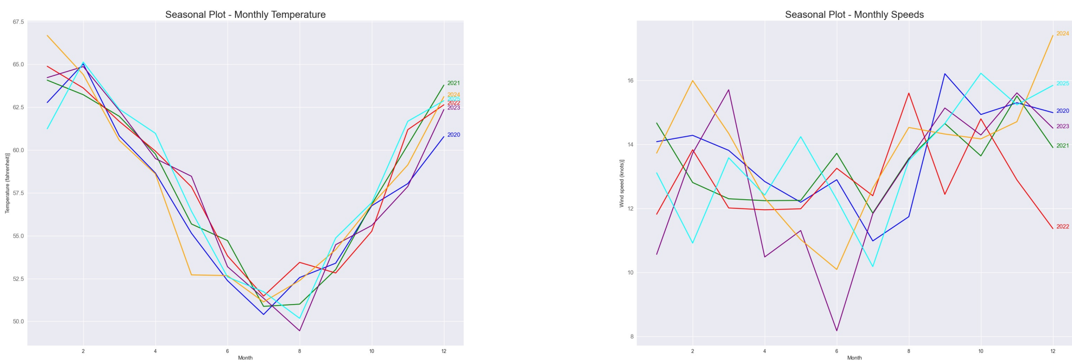

Figure 3.3: Seasonal patterns in temperature and wind speed at Wellington International Airport (2020-2025)

During the preprocessing steps we decided to drop columns that would be ineffectual to our models training. Columns such as Station Identifier were dropped as the experiment only concerns itself with the NZWN weather station for the international airport. Mean Sea Level Pressure also consisted of mostly missing values that could not be leveraged with imputation approaches as more than 90% of this data was missing. This was the same trend for variables pertaining to “Sky Conditions”, “Sky Level”, “Peak Wind Gust” and “Snow Depth”.

Deletion approaches were used as a preprocessing step for variables such as "Temperature", "Dew Point", "Relative Humidity", "Visibility".

Imputation approaches were utilised for Sky Cover as 30% of data had missing values entry. We used median approach that would help fill these missing columns with computed average of other observations.

Standardisation was also done with Python's StandardScaler which applies the Z-Score method for normalising the training dataset. This was done for most of the features in our data frame since they were of numerical type, apart from sky cover which was encoded using an Ordinal Encoder to preserve the natural order. For example, cloud cover is noted on METARs with:

- CLR $\rightarrow$ Clear
- FEW $\rightarrow$ Few
- SCT $\rightarrow$ Scattered
- BKN $\rightarrow$ Broken
- OVC $\rightarrow$ Overcast
- VV $\rightarrow$ Vertical Visibility (Visibility due to fog)

3.2.2 Feature Selection

Training our models on all variables can be costly and prone to overfitting where the model captures complex patterns during training creating complex decision boundaries that neglect generalisation on held out validation and test data. We want to make sure that only the variables that are the greatest predictors in contributing to the target are selected.

This feature selection process is done by leveraging scikit-learn's SelectKBest algorithm which trains a KNeighbors regressor on our METAR data and outputs the features that contribute the most to our target. We utilised a Pearson correlation to aid visually with the algorithm's decision. For example, when selecting the most informative features for temperature, a correlation matrix is shown for interactions between variables that contribute the most to temperature:

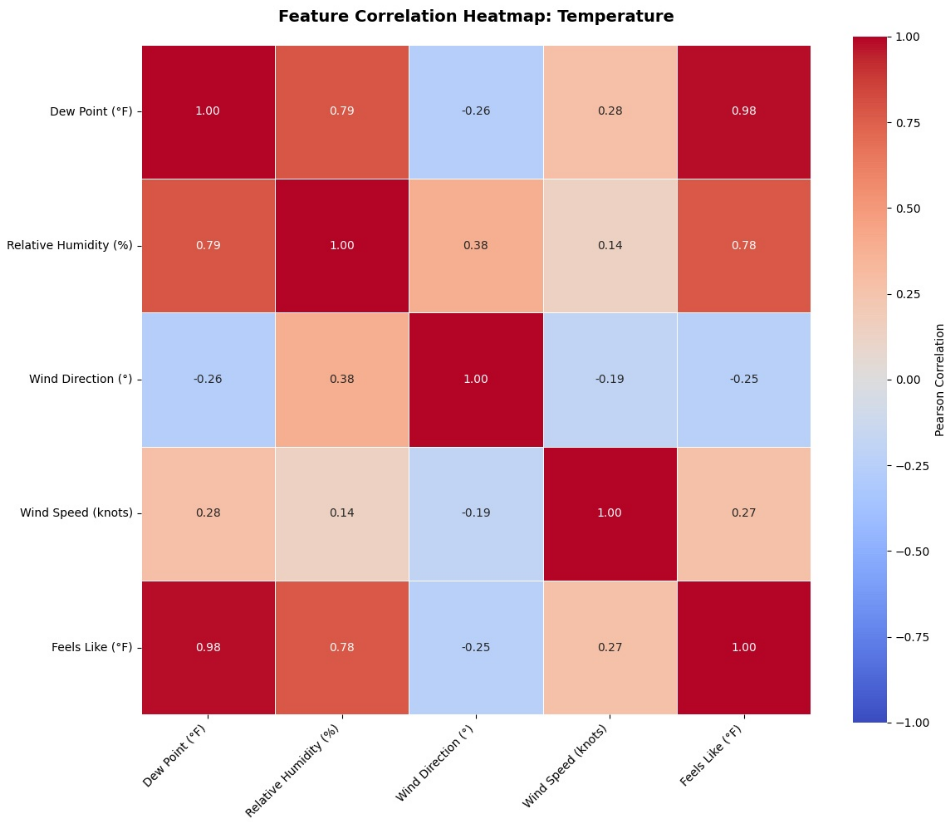

Figure 3.4: Feature Correlation Heatmap for Temperature

The map shows a strong positive correlation between Dew Point and "Feels Like" which makes intuitive sense since the dew point is a major driver of how temperature is perceived. Dew Point and Relative Humidity are also tightly linked at 0.79, same with Feel and Relative Humidity at 0.78. These variables clearly capture the overlapping information surrounding moisture in the air.

Seeing this visually helps us understand why the KBest algorithm selects "Relative Humidity", "Feel", "Dew Point", "Wind Direction" and "Wind Speed" as the top five features to be used during training because of feature selection.

This feature selection process and visual objection of correlations is repeated iteratively on our other targets that we are researching such as Cloud Cover, Visibility and Wind Speed.

Top Five Features Selected for Each Target

Wind Speed:

1. Sky Level
2. Temperature
3. Relative Humidity
4. Wind Direction
5. Feel

Visibility:

1. Sky Level
2. Relative Humidity
3. Wind Direction
4. Wind Speed
5. Sky Cover

Temperature:

1. Dew Point
2. Relative Humidity
3. Wind Direction
4. Wind Speed
5. Feel

3.3 Data Splitting

The entire dataset spans between 2020-2025 for Wellington Airport METAR reports. In machine learning, data is typically partitioned into training, validation and test sets. This partitioning enables evaluation of model performance and iterative adjustment of hyperparameters when generalisation performance is poor due to phenomena such as underfitting and overfitting.

The training data comprises instances spanning from 2020 to 2023. The validation set consists of instances from 2024. The testing set evaluates the model's ability to generalise on data from 2025. This yields an approximately 67/17/17 split in terms of percentage distribution.

3.4 Data Shifting

To evaluate the forecasting ability of deep learning models, it is necessary to predict a continuous value or classify a target within a future time interval. The entries in the dataset for Wellington International Airport include weather observations recorded every 30 minutes. This research examines the discriminative ability of deep learning approaches at a 1-hour forecasting interval to assess how weather changes relative to previous observation variables. This is accomplished by shifting the entire target column up by two positions to achieve the 1-hour prediction horizon.

Data entries at the bottom that contain NaN values after the shift operation are subsequently removed to avoid errors during the model training process.

3.5 Deep Learning Architectures

The experiments employ two types of deep learning models: a regular Artificial Neural Network using backpropagation and an LSTM (Long Short-Term Memory recurrent network). Previous literature demonstrates that when leveraging the temporal dynamics of time series data such as weather observations, an RNN with Long Short-Term memory capability can effectively capture temporal patterns in the data to produce more confident predictions or classifications [8].

This characteristic is particularly valuable for weather forecasting models as sequences can be constructed within the data over hourly periods, enabling the model to retain important trends while discarding uninformative data through the use of activation functions such as tanh and sigmoid.

3.5.1 Artificial Neural Network Architectures

To evaluate the impact of architectural complexity, both a baseline and extended model were implemented. This comparison enables assessment of how different target variables respond to varying architectural depths and determines whether increased complexity can outperform the LSTM architecture.

Baseline Model Architecture

- One Input Layer (Takes five input features from feature selection)
- One hidden layer scaled up with 64 hidden neurons
- One output Layer that predicts one continuous value (Regression)
- One output Layer that classifies between 8 categories (Classification)

Figure 3.5: Baseline ANN architecture parameters

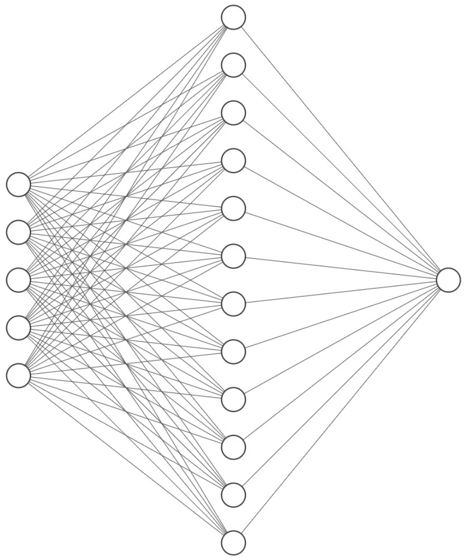

Figure 3.6: Baseline ANN architecture visualization

Extended Model Architecture

The extended model employs a funnel-like architecture that progressively reduces dimensionality from 5 input values to 64 channels, then 32, then 16, and finally to a single continuous value: $5 \rightarrow 64 \rightarrow 32 \rightarrow 16 \rightarrow 1$. This funnel-like architecture enables the model to leverage the inherent characteristics of meteorological variables such as wind speed, visibility and temperature. The architecture scales up to higher-dimensional feature spaces to learn complex relationships between variables, then progressively reduces dimensionality to retain only the most important relationships for generating accurate predictions.

- Input Layer: 5 Neurons
- Hidden Layer One: 64 neurons
- Hidden Layer Two: 32 neurons
- Hidden Layer Three: 16 neurons
- Output Layer: One Neuron (Prediction)

Figure 3.7: Extended ANN architecture parameters

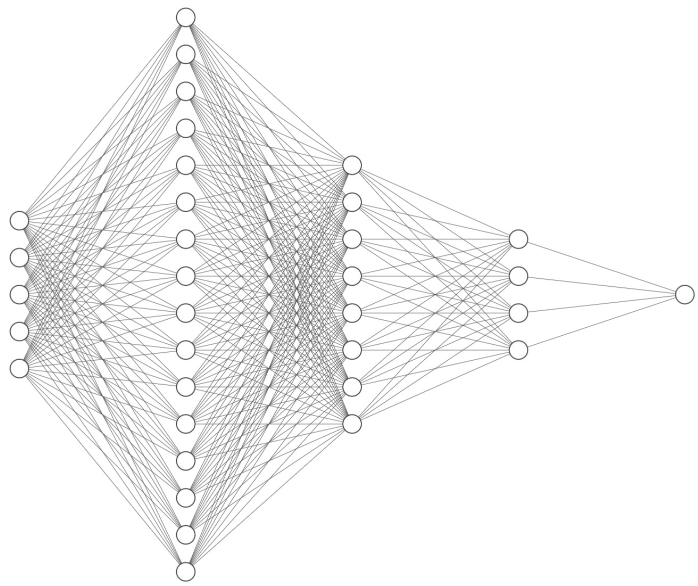

Figure 3.8: Extended ANN architecture visualization

Both models utilise the Mean Squared Error as the loss function:

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{3.1}$$

This loss function computes the difference between predicted and true labels, which is then used during backpropagation to update network gradients. This constitutes the learning process, which can be adjusted methodically through the learning rate. The models employ the Adam optimiser [12], which represents the most appropriate choice based on previous literature. Adam enables adaptive learning rates with faster training convergence compared to SGD (Stochastic Gradient Descent) and has minimal memory requirements. Each neural network uses the ReLU activation function between layers to compute the weighted sum, while the output layer utilises a Linear activation function. ReLU is selected for its ability to mitigate issues such as vanishing or exploding gradients [13], problems that frequently occur with other activation functions such as sigmoid and tanh.

Cloud Cover Classification Architecture

For cloud cover classification, a single neural network architecture was implemented to evaluate classification accuracy. This network incorporates dropout as a regularisation technique to prevent overfitting and batch normalisation for each layer to accelerate and stabilise training. The neural network architecture consists of an input layer, three hidden layers and one output layer.

- Input Layer: 5 Neurons
- Hidden Layer One: 128 neurons

- Hidden Layer Two: 64 neurons
- Hidden Layer Three: 32 neurons
- Output Layer: 8 neurons (8 categories of cloud cover types to classify)

Figure 3.9: Cloud Cover ANN architecture parameters

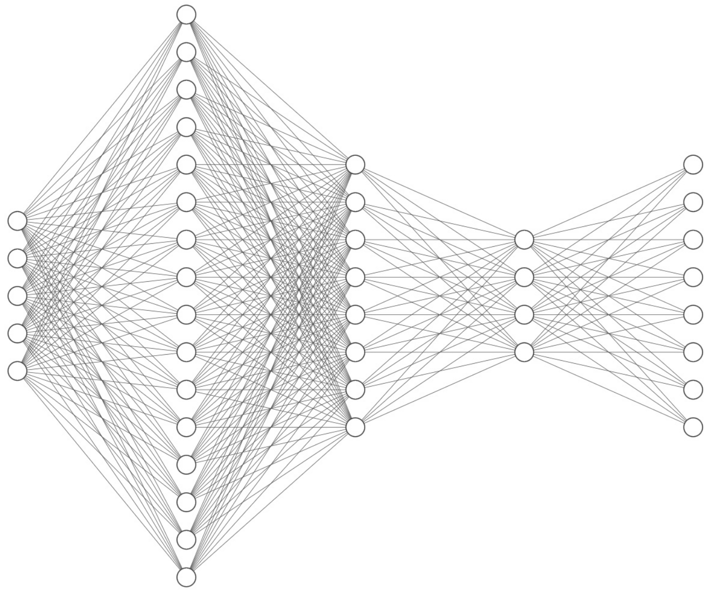

Figure 3.10: Cloud Cover ANN architecture visualization

3.5.2 LSTM Architecture

LSTMs are an extension of RNNs designed to address the vanishing gradient problem that plagues vanilla RNNs. The key innovation is the cell state, which acts as a highway for information to flow across time steps with minimal transformations.

The LSTM architecture consists of three main gates that regulate information flow:

1. **Forget Gate**: Decides what information to discard from the cell state using a sigmoid activation:
$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \tag{3.2}$$

2. **Input Gate**: Determines what new information to store in the cell state:
$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \tag{3.3}$$
$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \tag{3.4}$$

3. **Output Gate**: Controls what information from the cell state to output:
$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \tag{3.5}$$

The cell state is then updated as:

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \tag{3.6}$$

And the hidden state is:

$$h_t = o_t \odot \tanh(C_t) \tag{3.7}$$

This gating mechanism allows LSTMs to selectively retain or forget information over long sequences, making them particularly effective for time series forecasting where temporal dependencies are crucial.

3.6 Model Training

3.6.1 ANN: Base and Extended Model

The ANN is trained on data from 2020-2023 and validated using data from 2024 to monitor the differences between training/validation loss and training/validation MAE. The models are trained for 200 epochs.

3.6.2 LSTM

The LSTM is trained on data partitioned into sequences spanning 24-hour periods. This sequencing enables the LSTM to capture important temporal patterns when predicting targets such as temperature or wind speed. The proposed LSTM architecture consists of three hidden layers with 128 hidden neurons in each layer and incorporates a dropout parameter of 0.2 as a regularisation technique to mitigate overfitting. Data is processed in batches with a batch size of 32. The LSTM employs the Mean Squared Error loss function and Adam optimiser for learning, and trains for 100 epochs with early stopping implemented if validation MAE does not improve after a defined threshold.

3.7 Evaluation Metrics

3.7.1 Regression Metrics

The performance metrics we used to consider our models predictive ability for continuous values was the Mean Absolute Error, Mean Squared Error, and Root Mean Squared Error (MAE, MSE, RMSE).

Mean Squared Error calculates the square of errors between predicted and actual values:

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{3.8}$$

Mean Absolute Error calculates the average magnitude of errors between predicted and actual values:

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{3.9}$$

Root Mean Squared Error calculates the square root of the average squared differences between predicted and observed values:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{3.10}$$

3.7.2 Classification Metrics

For the cloud cover classification problem, the Cross Entropy Loss function was utilised to measure model performance by quantifying the difference between true labels and predicted probabilities:

$$\text{CrossEntropy} = -\sum_{i=1}^{C} y_i \log(\hat{y}_i) \tag{3.11}$$

For performance evaluation of the classification problem, Receiver Operating Characteristic (ROC) curves were employed to determine classifier performance across all threshold values. This analysis compares the relationship between the true positive rate (TPR) and false positive rate (FPR) to evaluate how effectively the model separates classes. The Area Under the Curve (AUC) measures overall performance, where an AUC of 1 represents a perfect model, while 0.5 indicates random guessing.

Confusion matrices were also employed to visualise how well predicted instances were correctly classified across the true labels. This matrix provides a visual representation of where the majority of labels are distributed among True Positives, False Positives, True Negatives, and False Negatives.

Chapter 4

Results

4.1 Artificial Neural Network Results

4.1.1 Wind Speed Prediction

The baseline ANN with only one hidden layer achieved a training MAE of 7.7517 knots and a validation MAE of 8.1250 knots over 200 training epochs for wind speed prediction.

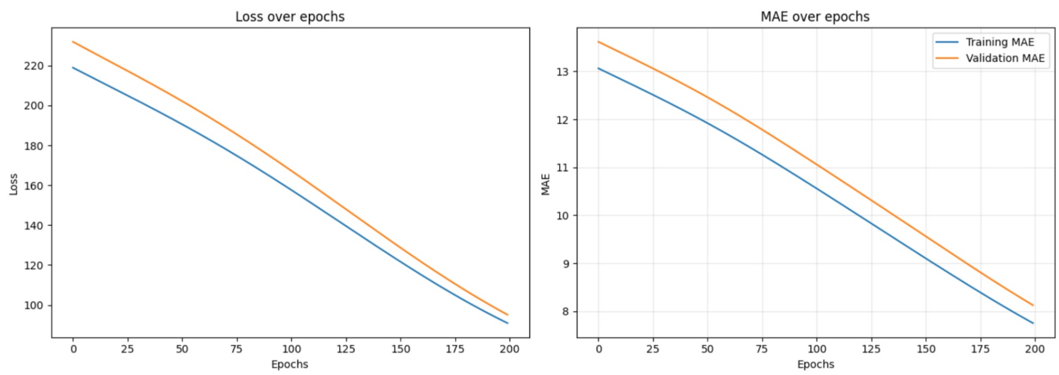

Figure 4.1: Training and Validation statistics for Baseline Model MAE and Loss (Wind Speed)

As shown in the figure, the model exhibits a linear decrease in both loss and MAE over the training epochs, rather than the desired steep initial descent that would indicate rapid learning and error reduction.

The extended ANN with three hidden layers achieved a training MAE of 5.4205 knots. The validation MAE was 5.2271 knots, representing a 2.8979 knots improvement compared to the baseline model with one hidden layer.

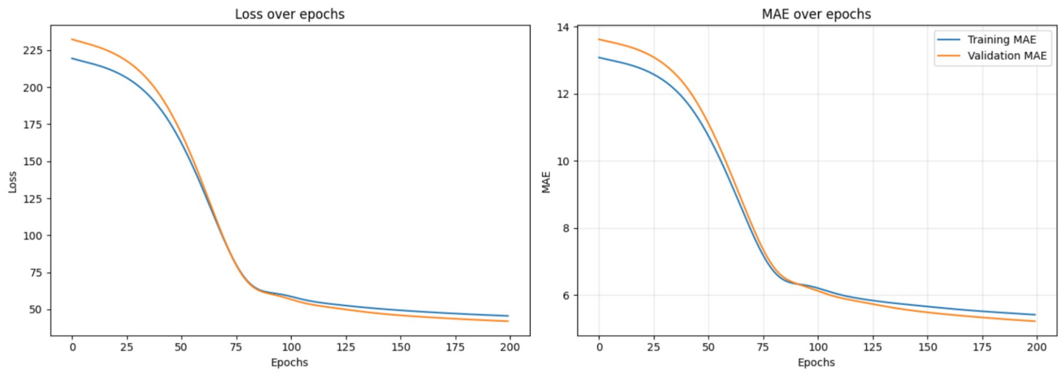

Figure 4.2: Training and Validation statistics for Extended Model MAE and Loss (Wind Speed)

The model exhibits initial learning difficulties before achieving a steeper descent at approximately 50 epochs, with the validation loss curve closely following the training curve in error reduction as the number of epochs increases.

4.1.2 Visibility Prediction

The baseline ANN with only one hidden layer had a training MAE of 1.4557 SM and a validation MAE of 1.3851 SM over 200 training epochs when considering visibility.

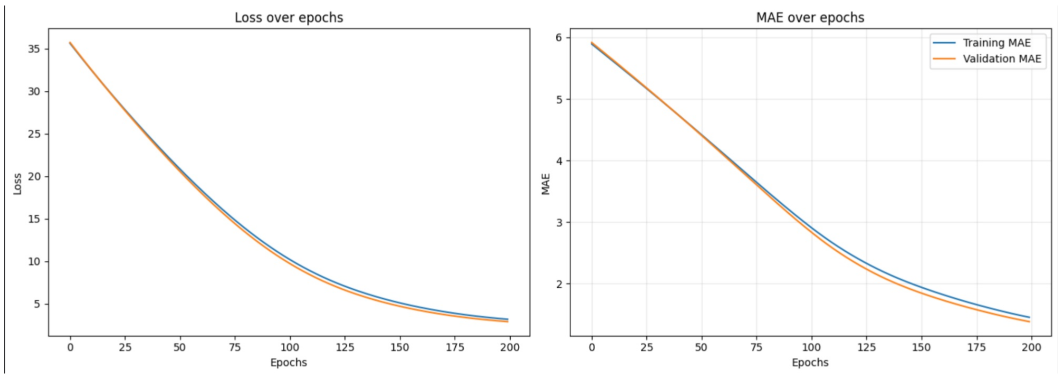

Figure 4.3: Training and Validation statistics for Baseline Model MAE and Loss (Visibility)

The extended ANN with three hidden layers had a training MAE of 0.6947 SM. The validation MAE was 0.6347 SM which is a 0.7504 SM improvement compared to the baseline model with one hidden layer.

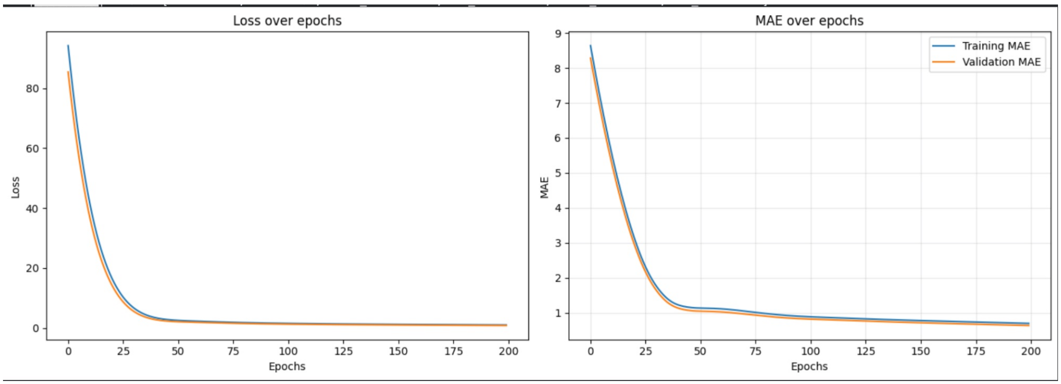

Figure 4.4: Training and Validation statistics for Extended Model MAE and Loss (Visibility)

4.1.3 Temperature Prediction

The baseline ANN with only one hidden layer had a training MAE of 50.4833°F and a validation MAE of 50.3507°F over 200 training epochs when considering temperature. The baseline model struggles to capture the required relationship with temperature during training.

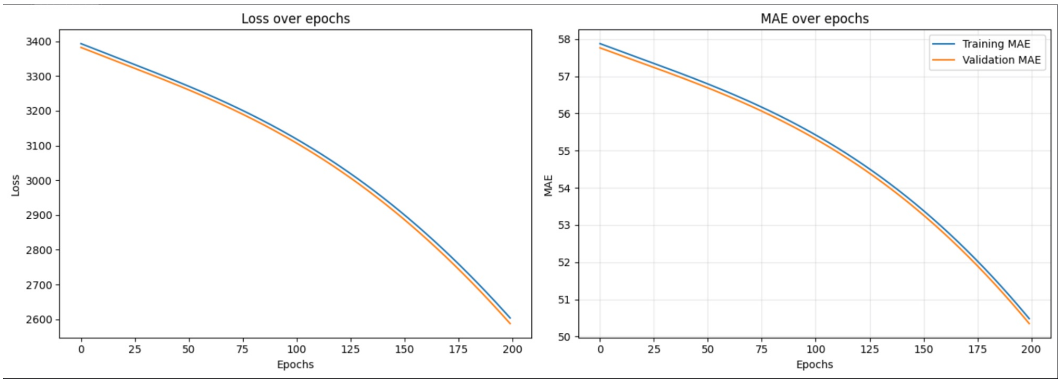

Figure 4.5: Training and Validation statistics for Base Model MAE and Loss (Temperature)

The extended ANN with three hidden layers had a training MAE of 6.0396°F. The validation MAE was 6.0396°F which is a considerable MAE improvement of 44.3111°F compared to the baseline model with one hidden layer.

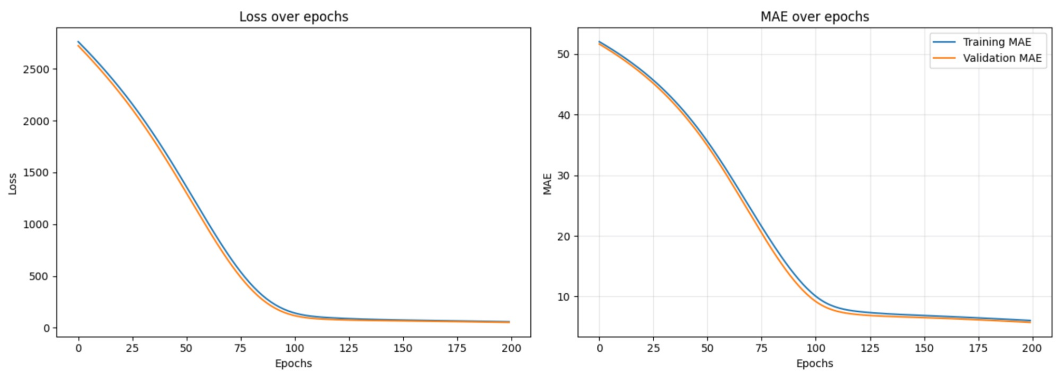

Figure 4.6: Training and Validation statistics for Extended Model MAE and Loss (Temperature)

4.1.4 Cloud Cover Classification

The Cloud Cover ANN had a training accuracy of 53.21% and validation accuracy of 52% over 200 training epochs when classifying outcomes of cloud cover such as “Broken”, “Scattered”, “Clear” etc.

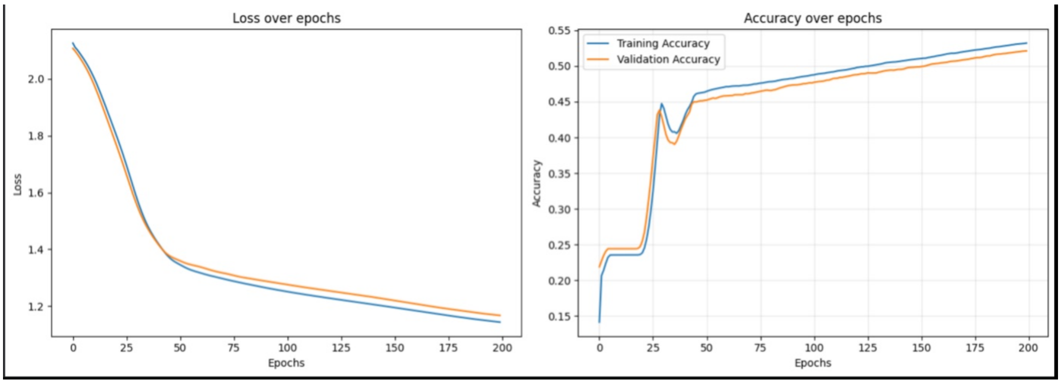

Figure 4.7: Training and Validation statistics for Extended Model MAE and Loss (Cloud Cover)

4.1.5 ANN Testing Results

Wind Speed Testing

When testing on data from 2025 at Wellington International Airport, the baseline ANN model demonstrated considerably better performance than the extended ANN model. The MAE for the extended model was 41.65 knots, whereas the MAE for the baseline model was 7.943 knots. This result is counterintuitive given that the training MAE and validation MAE indicated superior performance of the extended model. This discrepancy suggests that the learned representations may not transfer effectively to unseen data. These findings indicate that a simpler model architecture may generalise better for wind speed prediction.

Visibility Testing

A similar pattern was observed for visibility, where the ANN baseline model performed better compared to the extended version, achieving a baseline MAE of 1.44 SM compared to the extended MAE of 61.99 SM.

Temperature Testing

The extended ANN model demonstrated superior performance for temperature prediction, achieving an MAE of 6.0027°F while the baseline ANN model achieved an MAE of 50.54°F. This result demonstrates that the extended ANN model is more effective at capturing the complex decision space required to minimise prediction error for temperature. The funnel architecture enables the model to scale up to higher-dimensional feature spaces to learn important inter-variable relationships, then progressively reduce dimensionality to retain only the most critical relationships for accurate value prediction.

Table 4.1: ANN Baseline Model Testing Performance

Target	MAE	MSE	RMSE
Wind Speed	7.943 knots	96.092	9.802 knots
Visibility	1.443 SM	3.084	1.756 SM
Temperature	50.548°F	2608.384	51.072°F

Table 4.2: ANN Extended Model Testing Performance

Target	MAE	MSE	RMSE
Wind Speed	41.653 knots	1904.504	43.640 knots
Visibility	61.993 SM	4250.65	65.19 SM
Temperature	6.002°F	57.088	7.555°F

Cloud Cover Classification Testing

The testing accuracy for our ANN classifier was 54.084%. For a target that has 8 output categories to choose from is relatively impressive as it can be often hard to distinguish between categories that are similar in description. For example, the difference between few (FEW) and Scattered (SCT) is often small. This makes intuitive sense since 651 samples were predicted as FEW when the true label was SCT out of 2,965 samples, whereas the True Positive rate with SCT was 991/2,965 (33%).

ROC Curve:

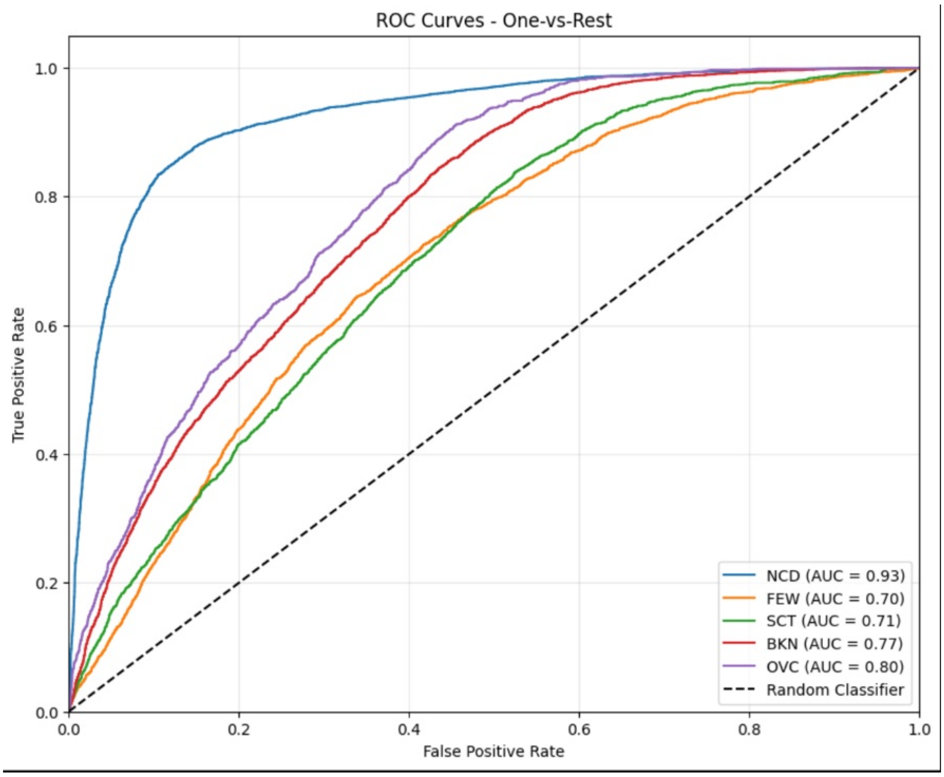

Figure 4.8: ROC Curves for ANN Cloud Cover Classification

The “No Cloud Detected” NCD category had the best AUC of 0.93 which shows that the classifier is great for classification thresholds considering no cloud. This makes sense since cloud can often take time to form which means that if the sky is clear then the chance of cloud category changing after 30 minutes or an hour is very small. “Overcast” OVC was the second best performing with an AUC of 0.8 which suggests an acceptable discriminative ability. An AUC of 0.5 suggests that the classifier is no better than random guessing. FEW and SCT both had AUC within 0.7 which is an acceptable discriminative ability, but as mentioned earlier has some discrepancies with the level of similarity between the two categories.

Confusion Matrix:

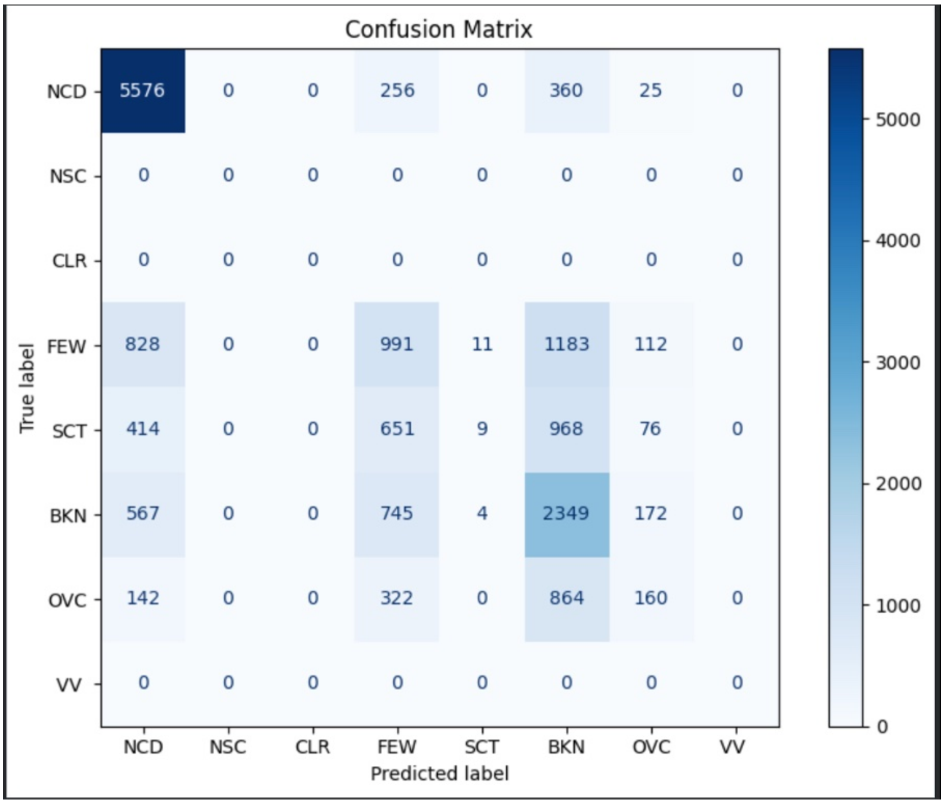

Figure 4.9: Confusion Matrix for ANN Cloud Cover Classification

We can visually see based on the concentration of correctly classified samples how NCD is the top performing for correctly classifying instances relating to no clouds being detected for 5576 samples out of 5576/7527 (74%). We can see for labels such as FEW and SCT that samples are evenly spread across the FEW, SCT, BKN and OVC categories. Although OVC has the second highest AUC a big portion of labels is misclassified as BKN instead of its true OVC label.

4.2 LSTM Results

4.2.1 Wind Speed Prediction

The LSTM had a training MAE of 1.914 knots and a validation MAE of 4.549 knots over 200 training epochs when considering wind speed. The LSTM outperforms both the base and extended ANN model. Where the LSTM beats the best ANN by 0.6781 knots when considering validation accuracy.

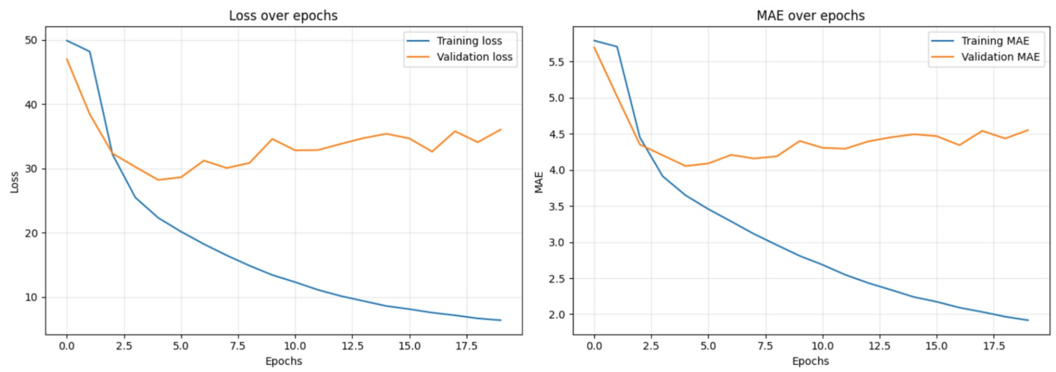

Figure 4.10: Training and Validation statistics for LSTM MAE and Loss (Wind Speed)

4.2.2 Visibility Prediction

The LSTM had a training MAE of 0.2381 SM and a validation MAE of 0.2864 SM over 200 training epochs when considering visibility. The LSTM outperforms both the base and extended ANN model. Where the LSTM beats the best ANN by 0.3483 SM when considering validation accuracy.

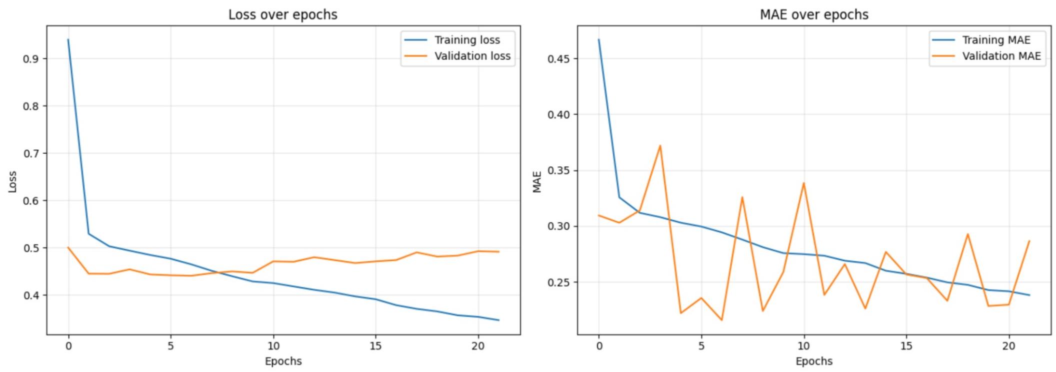

Figure 4.11: Training and Validation statistics for LSTM MAE and Loss (Visibility)

4.2.3 Temperature Prediction

The LSTM had a training MAE of 1.0140°F and a validation MAE of 1.2988°F over 200 training epochs when considering temperature. The LSTM outperforms both the base and extended ANN model. Where the LSTM beats the best ANN MAE by 4.7408°F when considering validation accuracy.

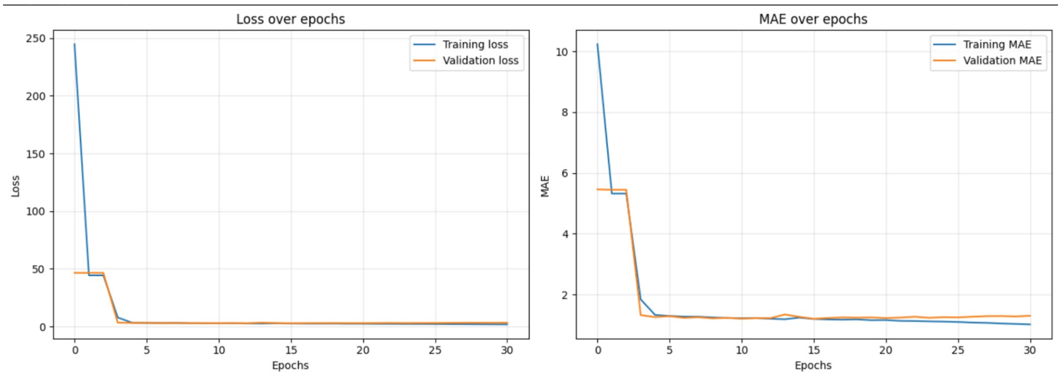

Figure 4.12: Training and Validation statistics for LSTM MAE and Loss (Temperature)

4.2.4 Cloud Cover Classification

The Cloud Cover LSTM had a training accuracy of 65.97% and validation accuracy of 52% over 200 training epochs when classifying outcomes of cloud cover such as "Broken", "Scattered", "Clear" etc. The validation accuracy has no difference to the original ANN model.

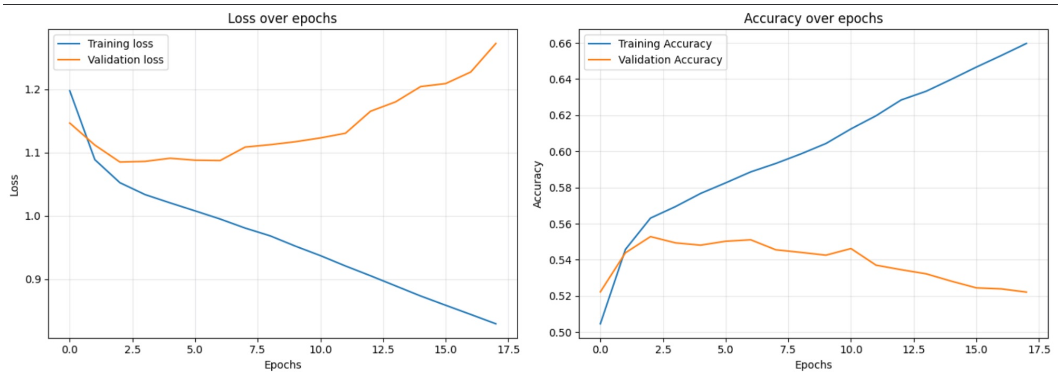

Figure 4.13: Training and Validation statistics for LSTM MAE and Loss (Cloud Cover)

4.2.5 LSTM Testing Results

Wind Speed Testing

When testing on data from 2025 at Wellington International Airport, the LSTM model demonstrated considerably superior performance compared to both base and extended ANN models. While the baseline ANN model achieved an MAE of 7.943 knots compared to the extended model, the LSTM achieved an MAE of 4.381 knots, representing a 3.112 knots improvement. This result demonstrates that the LSTM effectively leverages temporal patterns in sequential data to achieve robust generalisation on unseen data when forecasting wind speed at a 1-hour interval.

Visibility Testing

The LSTM model testing MAE for visibility was 0.28546 SM which is a 1.155 SM improvement compared to the better performing ANN with an MAE of 1.44 SM. The LSTM is the obvious deep learning approach when it comes to predicting a continuous value relating to visibility.

Temperature Testing

The LSTM model demonstrated superior performance when predicting temperature at a 1-hour interval. The testing MAE of 1.291°F represents a 4.711°F improvement compared to the better performing ANN with backpropagation. These results further validate the LSTM as the stronger deep learning approach for meteorological forecasting. The model effectively leverages sequential data over 24-hour periods at Wellington International Airport, enabling it to retain important characteristics within temporal patterns. This capability facilitates successful generalisation to new instances of temperature forecasted at a 1-hour interval.

Table 4.3: LSTM Model Testing Performance

Target	**MAE**	**MSE**	**RMSE**
Wind Speed	4.381 knots	35.721	5.977 knots
Visibility	0.285 SM	0.164	0.405 SM
Temperature	1.291°F	2.653	1.629°F

Cloud Cover Classification Testing

The testing accuracy for the LSTM Classifier was 54.326%. This represents a marginal difference compared to the ANN classifier accuracy of 54.084%. The difficulty of correctly classifying between 8 different categories is substantial; however, improvements on individual categories are evident through analysis of the ROC curve and confusion matrix.

ROC Curve:

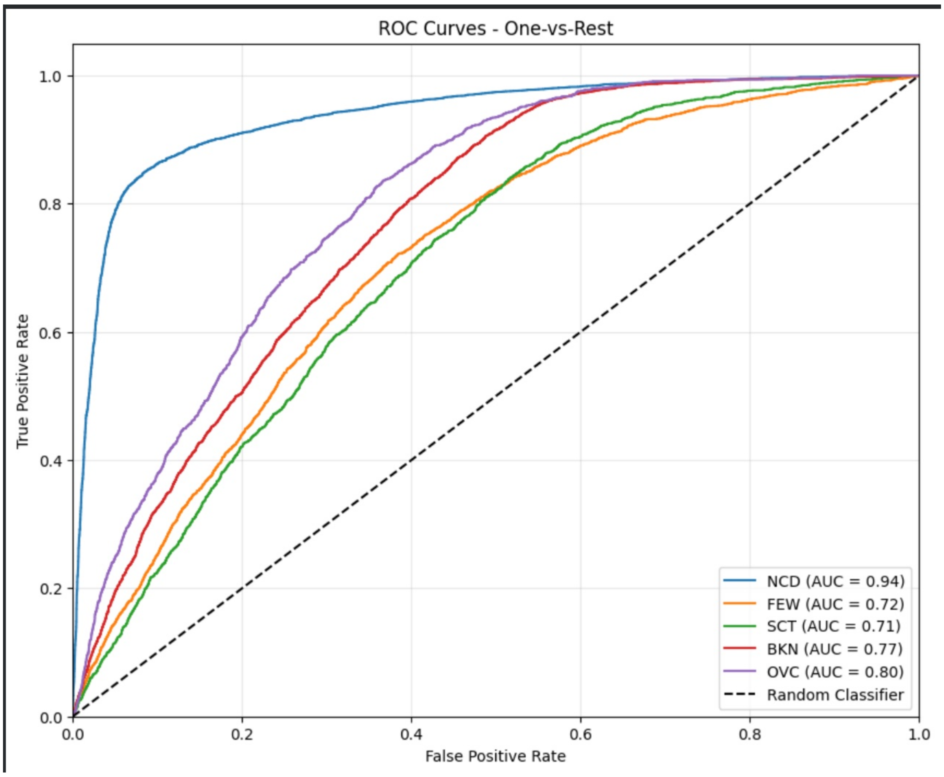

Figure 4.14: ROC Curves for LSTM Cloud Cover Classification

The AUC for NCD improved by 0.01, and FEW improved by 0.02 compared to the ANN classifier. Same relationship can be seen in this curve when also looking at the ROC curve for the ANN classifier.

Confusion Matrix:

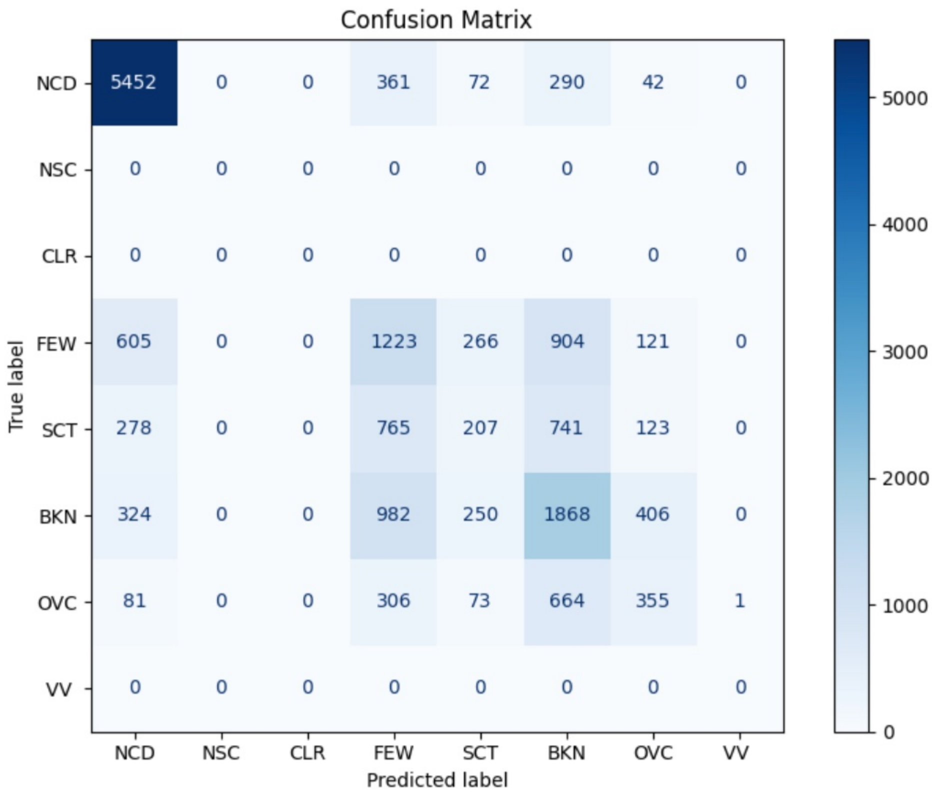

Figure 4.15: Confusion Matrix for LSTM Cloud Cover Classification

The True Positive Rate is slightly lower for the NCD category with the LSTM model, with 5452 correctly classified samples, whereas the ANN achieved 5576 samples. A similar pattern is observed for BKN, where the ANN demonstrates a higher TPR compared to the LSTM model.

Chapter 5

Discussion

When considering what deep learning approaches are most appropriate when predicting certain meteorological features that are necessary for safe flight operations, there are a lot of different trade-offs and context about which to use. For example, although we place importance on overall classifier accuracy, it is important to look at the true positive rate between labels, as pilots need to be assisted with the most up to date and correct weather information, as precision is not the trade-off used when faced with possible extreme weather events. Based on this logic, the ANN with backpropagation trained to classify labels relating to cloud cover is the better option for pilots.

For features that concern wind speed, visibility and temperature the LSTM is the clear-cut winner as its discriminative ability is excellent compared to our other deep learning approaches. LSTMs work well in leveraging the temporal discrepancies found in weather reports such as METARs which is why the architecture works well in producing clear generalisation on unseen METAR data to predict future forecasts at a 1-hour interval.

The experiments themselves ran smoothly and produced the results that we were after, however it would have been favourable to have paramount data quality as the raw data extracted from NZWN METAR reports had a considerable number of missing values and features that were essentially all NaN values. Some of these features could have also been helpful for some targets such as additional information pertaining to cloud cover and sky level.

It was particularly surprising to see our extended ANN model performing well on training data compared to our simpler baseline ANN but struggling to generalise well to unseen data. This is a common phenomenon of overfitting, where the model learns a complex decision boundary based off training data that fails to capture the general relationship. The extended model could have been too complex, with too many layers which is why the simpler version was more appropriate for the task to predict targets such as wind speed, visibility and temperature.

Chapter 6

Future Work

It would be interesting to see how the LSTM compares on a baseline to regular forecasting methods currently in place such as the TAF. This is the current weather technique in place as a form of short-term forecasting for pilots to consult. If the LSTM manages to have a lower propensity for error when it comes to classifying cloud cover or achieving a lower MAE compared to the TAF, then it would be a good talking point to see these models become operational. This could be gradually implemented across test sites such as small aerodromes with AFIS services to see the benefits. Some of these benefits could bring greater insight into flight planning, avoiding potentially dangerous flights and decreasing energy consumption for environmental benefit. Current weather stations utilise a lot of inference energy to forecast future conditions with current NWP methods, whereas deep learning models use a fraction of this energy.

For future experiments it would be favourable to have quality data available to help aid the experiment in achieving a better feature selection process before subsequently training models. It would also be interesting to look at maximum and minimum wind gust prediction, as Wellington International Airport tends to see events with considerable gusts throughout particular days during the calendar year.

Chapter 7

Conclusion

Air travel is projected to increase by a staggering 5.7%, where hundreds of millions of passengers travel each year [15]. Statistically, it calls to question when the next air traffic disaster could occur and not if it will occur. This is why it is incredibly important to have accurate weather predictions to help assists pilots in making safe flight planning decisions. Overall, our LSTM found itself to be the best performing approach when it comes to deep learning approaches that help forecast new weather conditions at a 1-hour interval. This 1-hour forecast could help airport operations run smoothly against delays when timely decisions are made about the safety of flights occurring. It gives pilots and cabin crew ample time to assess the forecasts and how best to prepare a flight in response to adversarial weather conditions to ensure safe flight operations.

Appendix A

Experimental Parameters

A.1 ANN Baseline Model

Table A.1: ANN Baseline Hyperparameters

Parameter	**Value**
Input Layer	5 neurons
Hidden Layer 1	64 neurons
Output Layer	1 neuron (regression) / 8 neurons (classification)
Activation Function (Hidden)	ReLU
Activation Function (Output)	Linear (regression) / Softmax (classification)
Loss Function	MSE (regression) / Cross-Entropy (classification)
Optimiser	Adam
Learning Rate	0.001
Batch Size	32
Epochs	200

A.2 ANN Extended Model

Table A.2: ANN Extended Hyperparameters

Parameter	Value
Input Layer	5 neurons
Hidden Layer 1	64 neurons
Hidden Layer 2	32 neurons
Hidden Layer 3	16 neurons
Output Layer	1 neuron (regression) / 8 neurons (classification)
Activation Function (Hidden)	ReLU
Activation Function (Output)	Linear (regression) / Softmax (classification)
Loss Function	MSE (regression) / Cross-Entropy (classification)
Optimiser	Adam
Learning Rate	0.001
Batch Size	32
Epochs	200

A.3 Cloud Cover ANN Model

Table A.3: Cloud Cover ANN Hyperparameters

Parameter	Value
Input Layer	5 neurons
Hidden Layer 1	128 neurons
Hidden Layer 2	64 neurons
Hidden Layer 3	32 neurons
Output Layer	8 neurons
Activation Function (Hidden)	ReLU
Activation Function (Output)	Softmax
Dropout	0.3
Batch Normalisation	Yes
Loss Function	Cross-Entropy
Optimiser	Adam
Learning Rate	0.001
Batch Size	32
Epochs	200

A.4 LSTM Model

Table A.4: LSTM Hyperparameters

Parameter	**Value**
Input Features	5
LSTM Layers	3
Hidden Units per Layer	128
Dropout	0.2
Sequence Length	24 hours
Output Layer	1 neuron (regression) / 8 neurons (classification)
Activation Function (LSTM)	tanh, sigmoid
Activation Function (Output)	Linear (regression) / Softmax (classification)
Loss Function	MSE (regression) / Cross-Entropy (classification)
Optimiser	Adam
Learning Rate	0.001
Batch Size	32
Epochs	100
Early Stopping Patience	10 epochs

A.5 Activation Functions

A.5.1 ReLU (Rectified Linear Unit)

$$f(x) = \max(0, x) \tag{A.1}$$

Used in hidden layers to introduce non-linearity while avoiding vanishing gradient problems.

A.5.2 Sigmoid

$$f(x) = \frac{1}{1 + e^{-x}} \tag{A.2}$$

Used in LSTM gates to control information flow (values between 0 and 1).

A.5.3 Tanh (Hyperbolic Tangent)

$$f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} \tag{A.3}$$

Used in LSTM cells to scale values between -1 and 1.

A.5.4 Softmax

$$f(x_i) = \frac{e^{x_i}}{\sum_{j=1}^{K} e^{x_j}} \tag{A.4}$$

Used for multi-class classification output to produce probability distributions.

A.5.5 Linear

$$f(x) = x \tag{A.5}$$

Used for regression output layers to predict continuous values.

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